FACE Prep CAMPUS

Course 3: Data Visualization and Dashboards with Excel and Cognos

Module 1: Introduction to Data Visualization

Key Concepts & Vocabulary

- **Data Visualization:** The graphical representation of information to provide insights and support decision-making.
- **Aesthetics:** The visual appeal of a chart (color, font, layout).
- **Information Density:** The amount of information displayed per unit of area.
- **Visual Encoding:** Mapping data values to visual properties like length, color, or shape.
- **Pre-attentive Attributes:** Visual properties (color, size, orientation) that the brain processes subconsciously in $< 250ms$.

Technical Deep-Dive

1. The Purpose of Visualization

Visualization serves three primary analytical functions:

1. **Exploration:** Finding hidden patterns or outliers (e.g., a scatter plot during EDA).
2. **Validation:** Verifying a hypothesis or a statistical model.
3. **Communication:** Sharing insights with stakeholders to drive action.

2. The Psychology of Perception (Gestalt Principles)

- **Proximity:** Objects close to each other are perceived as a group.
- **Similarity:** Objects that look alike are perceived as related.
- **Enclosure:** Using borders or shading to highlight a specific data group.

Module 2: Dashboarding Essentials in Excel

Key Concepts & Vocabulary

- **KPI (Key Performance Indicator):** A quantifiable measure used to evaluate success against targets.
- **Dashboard:** A single-page visual interface providing at-a-glance views of KPIs.
- **Slicer:** An interactive button that filters Pivot Tables and charts simultaneously.
- **Timeline:** A specific type of slicer used for filtering dates.

Technical Deep-Dive

1. Excel Dashboard Workflow (The 5-Step Process)

1. **Data Tab:** The raw, "ugly" data.
2. **Calculation Tab:** Hidden sheet where Pivot Tables and formulas reside.
3. **Dashboard Tab:** The clean, final visual layer for the user.
4. **Interactivity:** Connecting Slicers to multiple Pivot Tables using **Report Connections**.
5. **Protection:** Locking the sheet to prevent formula tampering.

Module 3: Introduction to IBM Cognos Analytics

Key Concepts & Vocabulary

- **Cognos Analytics:** IBM's enterprise-level BI platform using AI to automate visualization.
- **Data Module:** A web-based tool in Cognos to model and join different data sources.
- **Intent-based Authoring:** Cognos's ability to suggest charts based on a natural language query.

Technical Deep-Dive

1. Cognos vs. Excel

Feature	Excel	IBM Cognos
Scale	Up to 1M rows	Billions of rows
Automation	Manual Pivot Tables	AI-driven narratives
Security	File-level	Role-based (RBAC)

Module 4: Advanced Cognos Features and AI

Key Concepts & Vocabulary

- **AI Assistant:** A chat interface for questions like "Predict next month's churn."
- **Narratives:** Automatically generated text summaries explaining the "Why" behind charts.
- **Storytelling:** A feature creating a sequence of "scenes" with live data.

2. Best Practices for Professional Dashboards

- **The "F" Pattern:** Users scan in an "F" shape. Place key KPIs in the **Top Left**.
- **Consistency:** Use the same color for the same category across all charts.
- **Avoid 3D:** 3D charts distort data (Chart Junk).

Critical Exam Prep Summary Checklist

The "Logic Map" for Course 3 Quizzes

- **Difference between a Report and a Dashboard?** A report is detailed/historical; a Dashboard is high-level/real-time.
- **Function of a Slicer?** To provide a user-friendly way to filter Pivot Tables.
- **How does Cognos suggest a chart?** Via **Intent-based Authoring** or the **AI Assistant**.

Quantitative Benchmarks

1. **Human Vision:** We can distinguish between **5-7** different colors simultaneously.
2. **Latency:** Dashboards should load in under **3 seconds** to prevent a **40%** drop in user engagement.
3. **Color Blindness:** $\sim 8\%$ of males have color vision deficiency. Avoid Red/Green without secondary icons.

Reccomendations & Academic References

Recommended Books

- **"The Visual Display of Quantitative Information"** by Edward Tufte. The seminal text on the Data-Ink Ratio and avoiding "Chart Junk."
- **"Storytelling with Data: A Data Visualization Guide for Business Professionals"** by Cole Nussbaumer Knaflic. Essential for the "Communication" phase of the 5-step process.
- **"Information Dashboard Design"** by Stephen Few. Focuses on the cognitive psychology of rapid data consumption.

Reading References & Blog Citations

- **IBM Documentation:** Cognos Analytics with Watson Guide. Detail on "Intent-based Authoring" logic and data module joining.
- **Nielsen Norman Group:** The F-Shaped Scanning Pattern. Quantitative proof of user eye-tracking on digital dashboards.
- **Storytelling with Data Blog:** The 3-minute story. A guide for students on how to condense complex analysis for executive stakeholders.
- **Making data mean more through storytelling | Ben Wellington | TEDxBroadway :** Talk [Helps a lot in terms of improving delivery and presentation quality]